

TESTs - Lesson 18 - Gas limit and Loops

What was the platform, created by OpenZeppelin, we used in the previous lesson to demonstrate smart contract security challenges?

A Remix IDE

B Ethernaut

C Hardhat

In the Solidity v0.6.8 contract shown, what happens when a user with 20 tokens transfers 21 tokens?

A The transaction fails because of insufficient funds.

B An integer underflow occurs, resulting in the user having a massive number of tokens.

C The user's balance becomes -1.

What is the main reason for having a gas limit on blockchain transactions?

A To make transactions more expensive for users.

B To prevent a single transaction from using too many resources and blocking the network.

C To increase the speed of the blockchain.

What is the consequence of a transaction running out of gas before it completes?

A The transaction is partially executed and saved.

B All state changes are reverted, but the user still pays the gas fees.

C The transaction completes successfully but at a higher cost.

Why are unbounded loops considered dangerous in smart contracts?

A They can cause the contract to run out of gas, leading to transaction failure and potential Denial of Service (DoS).

B They make the code harder to read and maintain.

C They are not supported in newer versions of Solidity.

What is the "block gas limit"?

- A The maximum gas a single user can spend per day.
- B A limit set by the developer on how much gas a smart contract function can use.
- C A network-wide property that limits the total amount of gas for all transactions in a single block.

What is this technique called for handling large operations that would exceed the gas limit?

- A Gas Optimization
- B Amortization of Work
- C State Channeling

How does the "Amortization of Work" pattern help avoid hitting the gas limit?

- A It makes the code more efficient to use less gas overall.
- B It moves all computation from the EVM (scarce resource) to the client (abundant resource).
- C It breaks a large task into smaller steps that are executed across multiple transactions, tracking progress in state variables.